Lab # 11

Strings in C++

Objective:

The students will learn an apply the

- C-sting
- Declaration and Intilization of C string
- String Object

Introduction

A string is a sequence of characters used to store text. In C++, strings can be handled in two ways:

1. Using C-style strings (character arrays)
2. Using string objects (from the standard library)

Both methods are useful for storing and manipulating text.

C-Strings

C-strings are arrays of characters that end with a special character called the null character ('\0'). This character shows the end of the string.

Example

```
char name[] = "Ali";
```

Declaration and Initialization of C-Strings

C-strings can be declared and initialized in different ways:

```
char str1[10] = "Hello";
char str2[] = {'H', 'i', '\0'};
```

- The size of the array must be enough to store characters and the null character.
- The null character is automatically added when using double quotes.

String Object

C++ provides a built-in string class using the <string> library. It is easier to use than C-strings.

Example:

```
#include <string>
using namespace std;

string name = "Ali";
```

Features:

- No need to worry about size
- Easy to use functions like length, append, etc.

Conclusion

In this lab, we learned about different types of strings in C++. We understood how to declare and initialize C-strings and how to use string objects. String objects are easier and safer to use compared to C-strings.

Lab 11 Tasks

1. String Length Calculation

• Write a C++ program that declares and initializes a string variable called message with a phrase of your choice.

• Use a built-in function to calculate and display the length of the String

Code:

```
#include <iostream>
#include <string>  // string use karne ke liye library

using namespace std;

int main() {
    // ek string variable "message" declare aur initialize kiya
    string message = "Hello, this is my first C++ program";

    // length() function se string ki length nikal rahe hain
    int length = message.length();

    // result display kar rahe hain
    cout << "Message: " << message << endl;
    cout << "Length of the string: " << length << endl;

    return 0; // program successfully end
}
```

2. String Concatenation

▪ Declare two string variables called firstName and lastName and initialize them with your first and last name.

▪ Concatenate the two strings to form a full name and display it.

Code:

```
#include <iostream>
#include <string>  // string use karne ke liye library

using namespace std;

int main() {
    // do string variables declare aur initialize kiye
    string firstName = "Salar";
    string lastName = "Khan";

    // dono strings ko jor kar full name bana rahe hain
    string fullName = firstName + " " + lastName;

    // result display kar rahe hain
    cout << "First Name: " << firstName << endl;
    cout << "Last Name: " << lastName << endl;
    cout << "Full Name: " << fullName << endl;

    return 0; // program successfully end
}
```